VISUAL COMMUNICATION & VFX

ROTOSCOPY & KEYING

Frame-by-frame notes for the compositing artist

COURSE: Core – Rotoscopy & Keying (24BVS5CA)

PROGRAMME: B.Sc. Visual Communication and VFX | Semester V

COMPILED STUDY NOTES • ILLUSTRATED EDITION

Table of Contents

CHAPTER 01

The Origin of Rotoscopy & Its Importance in VFX

How a hand-traced animation trick from 1915 became the backbone of every modern visual-effects pipeline.

What is Rotoscopy?

Rotoscopy is the technique of tracing over motion-picture footage, frame by frame, to create a matte, an outline, or an animated element that is perfectly synchronised with the live-action plate underneath it. In today's digital pipeline, the "tracing" is done with vector shapes (splines) drawn directly on top of the video inside software such as Silhouette FX, Mocha Pro or Nuke, rather than with a pencil on paper. The output of this process — usually called a "roto matte" or simply "the roto" — is a black-and-white (or greyscale) mask that isolates a subject from its background so that compositors can replace the background, add effects behind the subject, or apply colour correction to only that subject.

The Origin — Max Fleischer's Rotoscope (1915)

The word "rotoscope" comes from the device invented by animator **Max Fleischer** and patented in 1917. Fleischer built a projector that cast one frame of live-action film at a time onto a glass panel or animation table. An artist would place a sheet of paper or animation cel over the projected frame and trace the outline of the actor's movement by hand. When every frame was traced and then played back in sequence, the result was an animation with startlingly fluid, life-like motion — motion that hand-drawn animation of the era could not match through pure guesswork.

Fleischer first used this technique for his "Out of the Inkwell" cartoon series, most famously to animate the character **Koko the Clown**. The realism of Koko's movement amazed audiences in an age when animation was still a young art form.

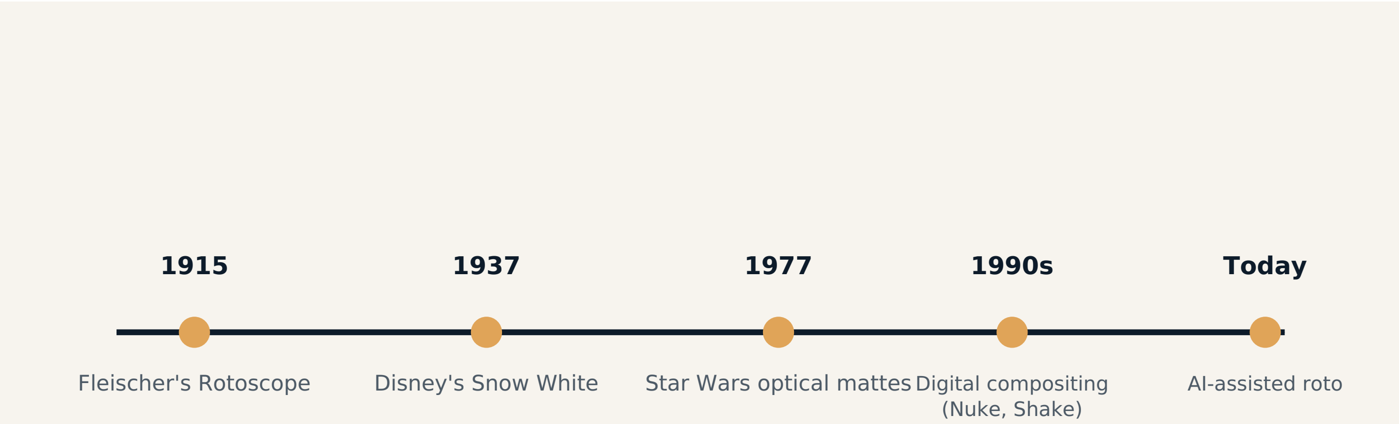

Fig 1.1 — Timeline showing how rotoscopy evolved from a hand-tracing device to an AI-assisted digital process.

From Analogue Tracing to Digital Masking

For most of the 20th century, rotoscopy remained a physical, hand-drawn process used mainly for animation reference (e.g., Disney's *Snow White and the Seven Dwarfs*, 1937) and for creating travelling mattes in optical film compositing — most notably the glowing lightsaber effects and space-ship mattes in the original *Star Wars* trilogy, where animators rotoscoped frame-by-frame mattes onto film stock by hand.

The arrival of digital compositing software in the 1990s (Nuke's predecessor at Digital Domain, Shake, Flame, Silhouette) replaced the pencil with the bezier spline. An artist now draws a vector shape on a monitor and keyframes its points across the timeline instead of physically drawing thousands of paper cels. The underlying goal, however, has not changed in over a hundred years: isolate a moving subject from the frame around it with pixel-accurate precision.

Why Rotoscopy Matters in Modern VFX

- **Background replacement** — Rotoscoped mattes let compositors swap out skies, cityscapes or entire environments behind an actor when a green screen was not used on set.
- **Object removal & clean-up** — Wires, rigs, crew members, camera reflections and modern-day objects in period films are removed by first rotoscoping around them.
- **Selective colour grading** — A colourist can isolate a character's jacket or a car with a roto shape and grade only that region without affecting the rest of the frame.
- **CG integration** — When a digital creature or vehicle must pass behind a real actor or object, roto shapes create the matte that lets the CG element tuck correctly behind foreground elements.
- **Beauty work & retouching** — Skin, blemish and continuity fixes in high-end commercials and films rely on precise rotoscoped regions.
- **Green-screen clean-up** — Even keyed footage often needs roto assistance around hair, motion blur, or areas where the key alone fails (spill, shadows, translucent objects).

WHY THIS STILL MATTERS IN THE AGE OF AI

Machine-learning rotoscopy tools (like Runway's auto-roto or Nuke's Copycat-trained models) can now automate a first pass, but a trained human artist is still required to clean up edges, handle motion blur, and maintain temporal consistency — the very skills this course builds.

Rotoscopy vs. Keying — Where They Differ

Keying (chroma-keying or luma-keying) automatically separates a subject from a background based on colour or brightness value — it works only when the plate was shot against a controlled background such as a green or blue screen. Rotoscopy, on the other hand, is a manual/vector-based process that works on *any* footage, including natural, uncontrolled backgrounds where no key is possible. In real productions, the two techniques are almost always combined: keying handles the bulk of a green-screen shot quickly, while rotoscopy is used to patch the areas the key gets wrong — fine hair, motion blur, shadows, and reflections.

WORKED EXAMPLE

In a superhero film, an actor is filmed on a green-screen stage swinging on a wire rig. The chroma key removes the green background instantly, but the thin steel wire and the actor's flyaway hair are not fully removed by the key. A roto artist draws a fine spline shape around the wire and hair strands for every frame of the shot to produce a clean, rig-free final composite.

KEY TAKEAWAYS

- Rotoscopy originated in 1915 with Max Fleischer's tracing device, used first for animation realism.
- The core principle — isolating a subject frame by frame — has remained unchanged for over a century.
- Digital rotoscopy uses vector splines instead of hand-drawn paper cels.
- Rotoscopy is essential for background replacement, clean-up, CG integration, and grading.
- Rotoscopy and keying are complementary, not competing, techniques in a real pipeline.

CHAPTER 02

Latest Tools Used in Rotoscopy

A tour of the industry-standard and emerging AI-assisted software that professional roto artists rely on today.

Why Tool Choice Matters

Every rotoscopy tool is built around the same core idea — drawing and animating vector shapes over footage — but each one optimises for a different part of the workflow: raw drawing speed, tracking accuracy, integration with a larger compositing pipeline, or automation through machine learning. A professional roto artist typically knows more than one tool because different studios and different shots call for different strengths.

Industry-Standard Roto & Tracking Software

1. Silhouette FX (Boris FX)

Silhouette is considered the gold standard for dedicated rotoscopy work. It offers extremely fast, precise spline tools, a powerful built-in tracker, multi-point tracking for stabilising shapes to moving footage, and a "Roto Paint" module for both rotoscoping and clean-up in a single application. Large VFX houses often route shots specifically into Silhouette for hair, fur and fine-edge rotoscopy because of its speed and its ability to handle thousands of points without slowing down.

2. Mocha Pro (Boris FX)

Mocha Pro is a planar-tracking powerhouse. Rather than tracking individual pixels, it tracks flat, planar surfaces (a wall, a phone screen, a face) using pattern and search regions. This makes it exceptionally good at rotoscoping objects that move, rotate or change perspective — the planar tracker can carry a roto shape through the whole shot with minimal manual keyframing, which is then refined by the artist.

3. Nuke — RotoPaint & Roto Nodes (Foundry)

Nuke is the node-based compositing application used across most major VFX studios. Its Roto and RotoPaint nodes allow artists to draw and animate shapes directly inside the same node graph used for the rest of the composite, which keeps everything — tracking, keying, roto, colour, and 3D integration — in one pipeline. Nuke's roto tools are typically used when the roto work must feed directly into a larger multi-artist composite.

4. Adobe After Effects (Roto Brush 2 & Pen Tool)

After Effects is widely used in smaller studios, motion graphics houses, and educational settings. Its **Roto Brush 2** tool uses a machine-learning segmentation engine — the artist paints a rough stroke on the foreground and background of one frame, and the software propagates an automatic matte across the rest of the shot, which the artist then refines. The classic Pen tool is still used for precise, manual bezier rotoscopy.

5. Blackmagic Fusion / DaVinci Resolve (Fusion page)

Fusion's node-based Polygon and B-Spline tools are common in broadcast and episodic television pipelines, especially now that Fusion ships free inside DaVinci Resolve, making it a popular entry point for students and freelancers.

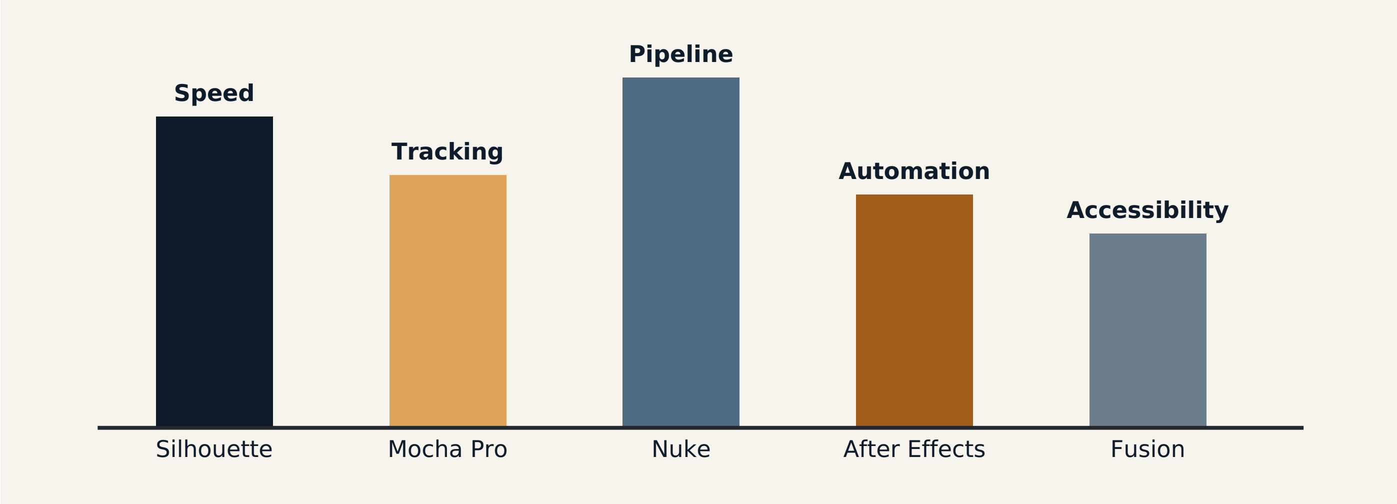

Fig 2.1 — Relative studio-perceived strength of each major roto tool (illustrative comparison).

Emerging AI-Assisted Rotoscopy Tools

The newest generation of rotoscopy tools uses trained neural networks to generate an initial matte automatically, dramatically speeding up the first pass of a shot:

- **Runway ML (Green Screen / Auto Roto)** — Web-based automatic subject segmentation that produces a rough matte in seconds from a single click, popular for quick social-media and indie work.
- **Nuke's Copycat / BlinkScript ML nodes** — Allow studios to train a custom machine-learning model on a handful of manually rotoscoped frames, then apply that trained model to automatically roto the rest of a long sequence.
- **Adobe Sensei (inside Roto Brush 2)** — The AI segmentation engine that powers After Effects' automatic subject tracking.
- **Segment Anything-style models** — General-purpose image segmentation networks increasingly integrated into compositing plug-ins to generate starting mattes for any object in a frame.

IMPORTANT DISTINCTION

AI tools are best understood as accelerators for the *first pass*, not replacements for an artist. Automated mattes still drift on hair, transparency, motion blur and complex overlaps — a trained rotoscopy artist is required to clean, track, and finish the shot to broadcast/film quality.

WORKED EXAMPLE

A VFX facility receives 400 shots of a crowd scene that need clean plates. Junior artists first run Roto Brush 2 in After Effects to get an 80% accurate matte on every shot within a day. Senior artists then open only the problem shots in Silhouette to hand-correct edges around hair and fast motion — cutting total turnaround time by more than half compared to fully manual rotoscopy.

TOOL	BEST FOR	KEY STRENGTH
Silhouette FX	Dedicated, high-end roto shots	Speed & precision with thousands of points
Mocha Pro	Moving/rotating planar objects	Planar tracking, minimal manual keyframes
Nuke RotoPaint	Studio pipeline integration	Node-based, works inside full composite
After Effects	Fast turnaround, motion graphics	Roto Brush 2 AI segmentation
Fusion / Resolve	Broadcast, freelance, students	Free, node-based, accessible

KEY TAKEAWAYS

- Silhouette, Mocha, Nuke, After Effects and Fusion are the five core roto tools used across the industry.
- Each tool is chosen for a specific strength: precision, planar tracking, pipeline integration, automation, or accessibility.
- AI-assisted rotoscopy (Roto Brush 2, Runway, Copycat) speeds up the first pass but still needs artist refinement.

- Professional artists typically combine two or more tools depending on the shot's needs.

CHAPTER 03

The GUI & Important Terminologies in Rotoscopy

Learning to read a rotoscopy workspace fluently — the panels, tools and vocabulary every artist must know.

Anatomy of a Rotoscopy Workspace

Although Silhouette, Nuke and After Effects each have their own layout, almost every rotoscopy application is built from the same five functional zones. Learning to recognise them lets an artist pick up a new tool quickly.

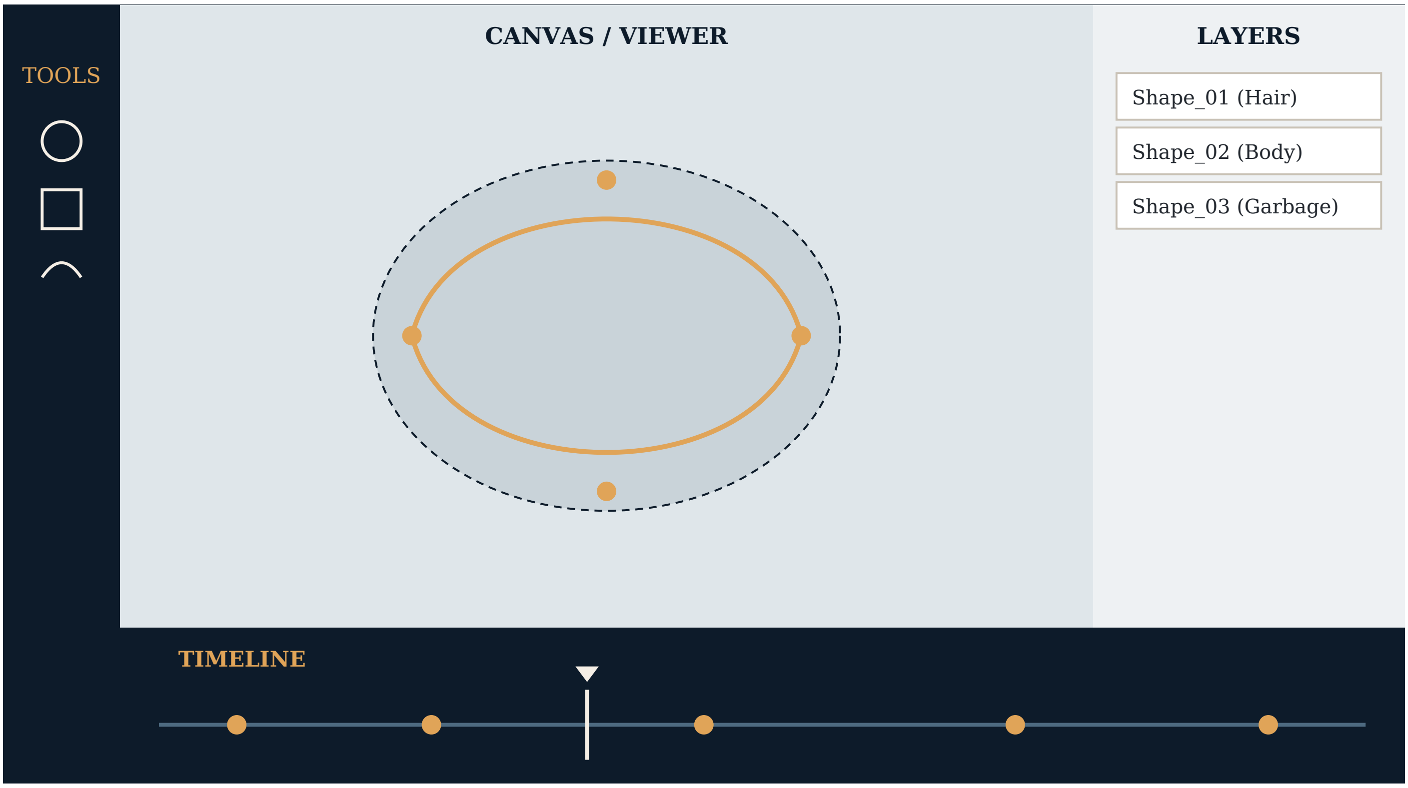

Fig 3.1 — Generic rotoscopy GUI layout: toolbar (left), canvas/viewer (centre), layers panel (right), timeline with keyframes (bottom).

1. The Toolbar

Contains the point/shape creation tools — Bezier pen, B-spline tool, X-spline tool, freehand draw, and the tracking/transform tools used to move or stabilise a shape.

2. The Canvas / Viewer

The main working area where the footage is displayed and where the artist places, edits and animates spline points directly on top of the image.

3. The Layers / Shapes Panel

Lists every shape drawn in the project — each with its own name, visibility toggle, blend mode and stacking order, similar to layers in Photoshop.

4. The Timeline

Shows the current frame position and every keyframe set on the selected shape, allowing the artist to scrub through the shot and check how the roto tracks the motion.

5. The Properties / Parameters Panel

(Not shown above for space, but present in most tools) — displays numeric values for feather, opacity, blend mode and motion-blur settings for the selected shape.

Essential Terminology

TERM	MEANING
Spline	A smooth vector curve made of connected points, used to draw the roto shape.
Point / Vertex	An individual control point on a spline that can be moved to reshape the curve.
Bezier Handle / Tangent	A directional handle attached to a point that controls the curvature on either side of it.
Matte	The black-and-white mask generated by a shape, used to isolate a subject.
Garbage Matte	A rough, loosely-drawn shape used to quickly exclude obviously unwanted areas before refining the key.
Core Matte	A tightly-drawn shape covering only the solid centre of a subject, used to guarantee full opacity in that region.
Feather	A soft, gradual falloff added to the edge of a shape so the matte blends naturally instead of looking hard-cut.
Keyframe	A frame where the shape's position/points are explicitly recorded; the software interpolates the frames between two keyframes.
Tracking	The automatic process of following a point or region's movement across frames.
Motion Blur (on a roto shape)	A setting that lets the matte edge blur to match the natural motion blur of the subject in the plate.
Interpolation	The mathematical calculation the software uses to fill in shape positions between two keyframes.

WORKED EXAMPLE

A student opens Silhouette for the first time. They select the Bezier tool from the toolbar, click four points around an actor's silhouette in the canvas, name the resulting shape "Actor_Roto" in the layers panel, then move to frame 24 in the timeline and nudge the points to match the actor's new pose — this automatically sets a keyframe.

KEY TAKEAWAYS

- Every roto GUI shares five zones: toolbar, canvas, layers panel, timeline, and properties panel.
- Splines are made of points and bezier handles that define the shape's curvature.
- Garbage mattes are rough and fast; core mattes are tight and precise — both are used together.
- Feather, keyframe, tracking and interpolation are the four terms used most often in daily roto work.

CHAPTER 04

Shapes & Paths Used in Rotoscopy

Bezier, B-spline and X-spline — the three families of curves every roto artist chooses between.

Why Shape Type Matters

Every rotoscopy shape is a closed (or open) path built from a series of points. How the software mathematically calculates the curve *between* those points determines how easy the shape is to control, how naturally it hugs an organic edge, and how quickly it can be built. This is why professional tools offer more than one shape type.

1. Bezier Shapes

The Bezier curve is the most common shape type, built from anchor points and adjustable tangent handles. Dragging a handle changes the curvature on both sides of the point (or just one side, if the handles are "broken"). Bezier curves give an artist precise, predictable control and are ideal for hard-edged or geometric subjects — a car door, a table edge, a building.

2. B-Spline Shapes

A B-spline (Basis spline) creates a smooth curve that passes *near* its points rather than directly through them, similar to a curve pulled taut by magnets at each point. This makes B-splines extremely fast for rotoscoping soft, organic shapes such as skin, cloth folds or hair clumps, because the artist needs far fewer points to get a smooth result compared to Bezier.

3. X-Spline Shapes

The X-spline (used prominently in Silhouette) is a hybrid: each point carries a "tension" value that can be adjusted to behave like a sharp Bezier corner at one moment and a smooth B-spline curve the next — without needing separate handle controls. This flexibility makes X-splines a favourite for complex organic outlines that have both sharp and soft sections, like a hand with fingers (sharp between fingers, soft along the back of the hand).

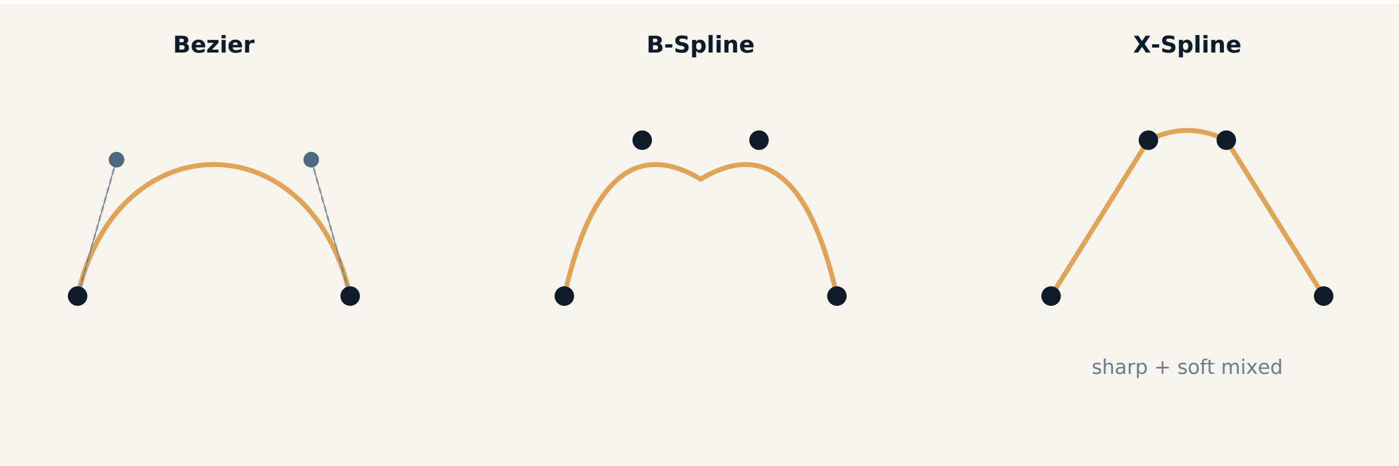

Fig 4.1 — Comparing curve behaviour: Bezier uses handles, B-spline glides near points, X-spline mixes sharp and soft per point.

Open Paths vs. Closed Shapes

A **closed shape** connects its last point back to its first, forming an enclosed region used as a matte (e.g., an outline around a person). An **open path** does not close, and is typically used for tasks like paint-fixes, wire-removal strokes, or guide lines that do not need to fill an area — only to mark a line.

WORKED EXAMPLE

A roto artist rotoscoping a woman's flowing hair uses an X-spline for the hairline (mixing sharp flyaway strands with a soft scalp curve), a Bezier shape for her rigid sunglasses frame, and an open B-spline path to mark a thin microphone wire for removal.

KEY TAKEAWAYS

- Bezier shapes use handle-based control — best for hard, geometric edges.
- B-splines glide near their points — fast for soft, organic shapes.
- X-splines mix sharp and soft behaviour per point — ideal for complex, mixed-edge subjects.
- Closed shapes create fillable mattes; open paths are used for lines, wires, and guides.

CHAPTER 05

The Process of Creating Splines

A step-by-step walkthrough of building a clean, efficient roto shape from the very first point.

Planning Before Drawing

Before placing a single point, an experienced roto artist studies the shot: Where does the subject move fastest? Where does it overlap other objects? Where is the edge sharp, and where is it soft or motion-blurred? This planning determines how many points are needed and where they should sit.

1. **Choose the reference frame.** Start on a frame where the subject's outline is clearest and least blurred — usually the middle of the shot, not the first frame.
2. **Place points at extrema.** Click points at the outermost/extreme edges of the shape first (top of the head, tips of shoulders, outer edge of hands) before filling in detail.
3. **Add points at direction changes.** Insert a point wherever the outline changes direction sharply — this is where a spline needs anchoring, not in the middle of a smooth, straight edge.
4. **Adjust tangents/handles.** Pull Bezier handles (or tension values on an X-spline) so the curve hugs the actual edge of the subject, not just a rough approximation.
5. **Close the shape.** Connect the final point back to the first to complete a fillable, matte-generating shape.
6. **Track/animate across frames.** Use the built-in tracker (or manual keyframing) to carry the shape through the shot, correcting drift every few frames.
7. **Refine & add feather.** Once the shape tracks correctly, add a soft feather to the edge so the final matte blends naturally into the new background.

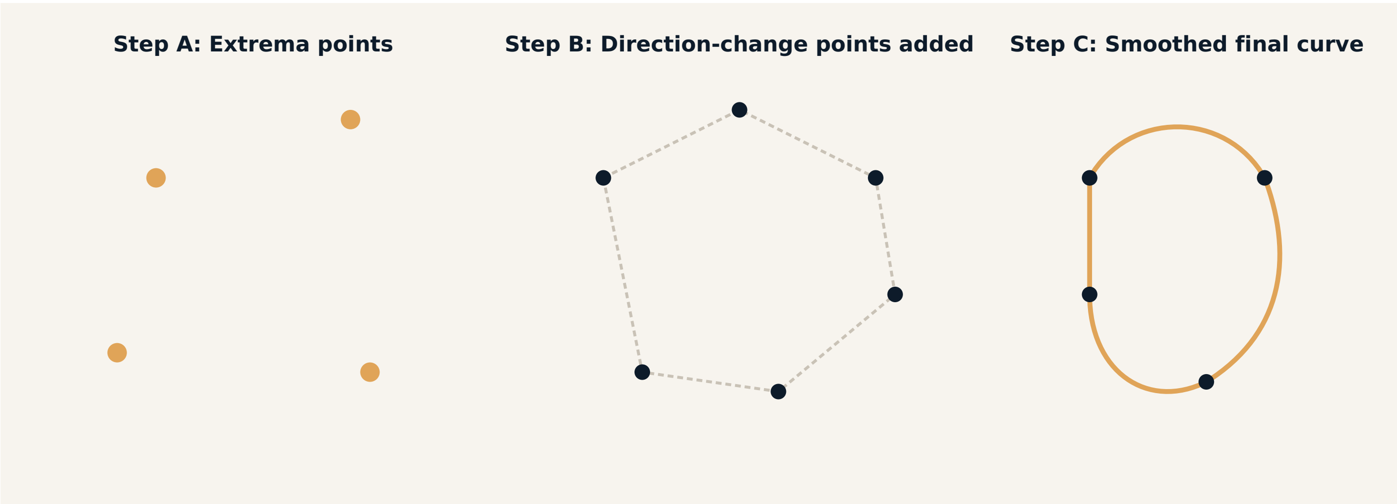

Fig 5.1 — Building a spline: mark extrema first, add direction-change points, then smooth the final curve with handles.

The "Minimum Points" Principle

A common beginner mistake is placing far too many points, which makes the shape stiff, slow to animate, and prone to wobbling ("chattering") between frames. The professional rule of thumb is: use the fewest possible points that still describe the shape accurately, and let the curve's natural bend do the rest of the work between points.

> **RULE OF THUMB**
> If you can remove a point and the curve still matches the edge within a pixel or two, remove it. Fewer points mean faster tracking, easier keyframe management, and smoother interpolation across frames.

WORKED EXAMPLE

A student rotoscoping a moving car door uses only six points — one at each corner and one at the curved top edge — instead of twenty points scattered randomly along the outline. The six-point shape tracks the door's motion smoothly across 120 frames with far fewer manual corrections than a busy, over-pointed shape would need.

KEY TAKEAWAYS

- Always start on the clearest reference frame, not necessarily frame 1.
- Place points at extrema and direction changes first — not evenly around the shape.
- Adjust tangent handles to hug the true edge before closing the shape.
- Fewer, well-placed points create a faster, smoother, more stable spline.

CHAPTER 06

Edge Consistency in Rotoscopy

Why a wobbling, inconsistent matte edge is the fastest way to break the illusion of a composite.

What is Edge Consistency?

Edge consistency refers to how stable and believable a roto matte's edge remains from one frame to the next. A matte edge that is tight in one frame and loose in the next — even by a pixel or two — creates a visible "wobble" or "chatter" when the sequence is played back. Because the human eye is extremely sensitive to motion and edges, this kind of inconsistency is one of the fastest ways to break the illusion that a composite is real.

Common Causes of Inconsistent Edges

- **Too many points** — Excess points shift independently frame to frame, causing micro-wobbles along the outline.
- **Poor tracking** — When automatic tracking loses the edge (due to motion blur, low contrast, or occlusion), the shape drifts off the real subject.
- **Inconsistent feather values** — Changing the feather amount between keyframes causes the edge softness to visibly pulse.
- **Ignoring motion blur** — A perfectly sharp roto edge on a fast-moving, naturally blurred subject looks artificially "cut out."
- **Uneven keyframe spacing** — Placing keyframes too far apart forces the software to guess (interpolate) the shape across too many frames, producing sliding or lagging edges.

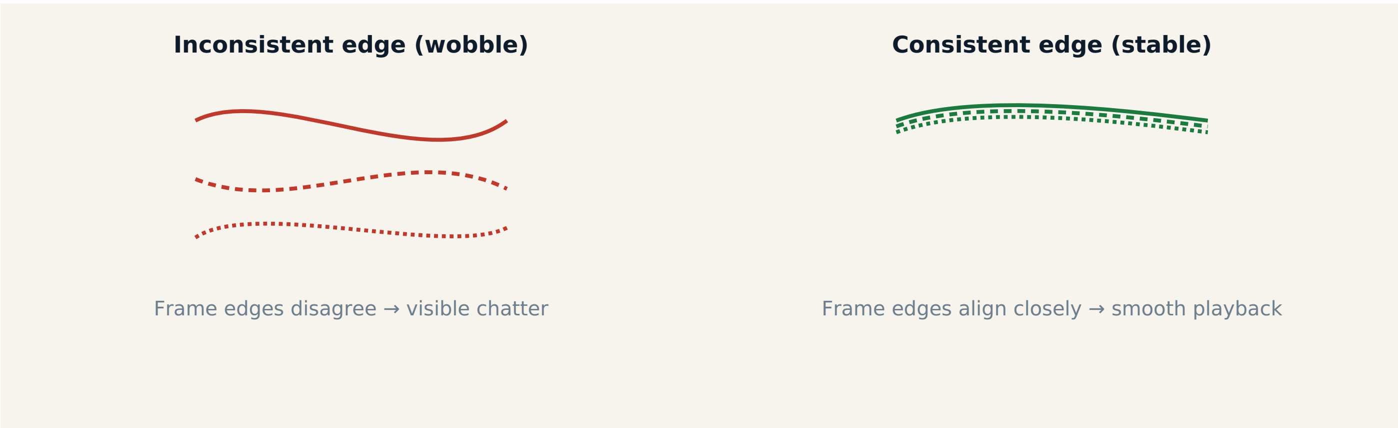

Fig 6.1 — Overlaying several frames of the same edge: an inconsistent roto wobbles between frames, while a stable roto stays aligned.

Techniques for Maintaining Edge Consistency

1. Lock Point Count Early

Decide the final point count on the clearest frame and avoid adding or deleting points later in the shot unless absolutely necessary — every added point is a new source of potential drift.

2. Use the Tracker, Then Verify by Eye

Let the built-in point or planar tracker carry the shape automatically, but scrub through every few frames to visually confirm the edge is still sitting correctly, correcting drift immediately rather than letting it accumulate.

3. Keep Feather Values Locked or Smoothly Animated

If feather must change over the course of a shot (e.g., subject moves from sharp focus into soft focus), animate it gradually across many frames rather than jumping the value suddenly at one keyframe.

4. Match Motion Blur to the Plate

Enable the shape's motion-blur setting and match its amount/angle to the visible blur in the footage so the matte edge does not look unnaturally crisp against a blurred subject.

5. Review in Motion, Not Just Frame-by-Frame

A roto shape can look perfect on a single paused frame yet wobble badly in playback. Always scrub and play the shot at speed before calling a roto "final."

WORKED EXAMPLE

An artist rotoscoping a runner notices the matte edge along the runner's leg appears to "breathe" in and out during fast playback, even though each individual frame looks fine when paused. Reviewing the shape's point count, they find two extra points were added mid-shot to chase a shadow — removing them and re-tracking with the original point count solves the wobble.

KEY TAKEAWAYS

- Edge consistency means the matte edge behaves believably and steadily across the whole shot, not just on one frame.
- Extra points, poor tracking, uneven feather and ignored motion blur are the top causes of inconsistency.
- Lock the point count early, verify tracking by eye, and animate feather smoothly.
- Always judge a roto shape in playback, not only frame-by-frame.

CHAPTER 07

Positive Space & Motion Paths

Reading the silhouette correctly and understanding the trajectory a shape must follow through a shot.

What is Positive Space?

In visual design and rotoscopy alike, **positive space** is the area occupied by the actual subject — the person, object or element being rotoscoped. **Negative space** is everything around it — the background, the empty gaps between limbs, or the visible sky between an arm and the body. A roto shape's job is to precisely and exclusively cover the positive space, leaving the negative space untouched.

Correctly reading positive space becomes especially important when a subject's silhouette has interior "holes" — for example, the negative space visible between a person's arm and their torso when a hand rests on a hip. An accurate roto must cut that negative space out of the matte, or the composite will show a background element bleeding incorrectly through a gap that should be empty.

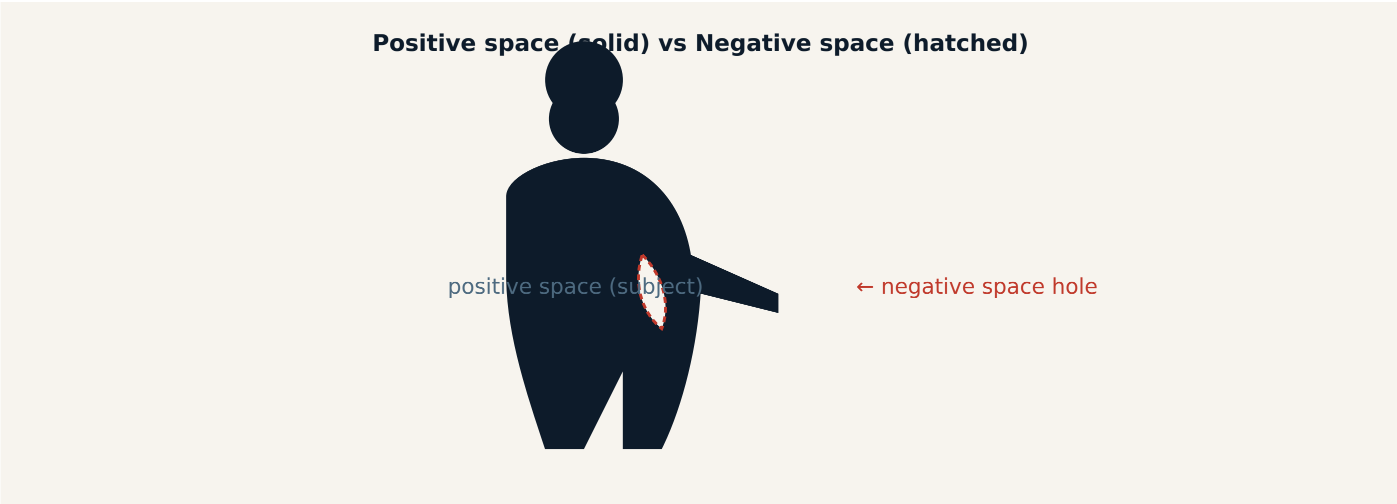

Fig 7.1 — The solid silhouette is positive space; the small hatched gap between the arm and body is negative space that must be cut out of the matte.

What is a Motion Path?

A motion path is the trajectory that a tracked point, shape, or object follows as it moves through a shot over time — visualised as a line connecting its position at every frame. Understanding the motion path of a subject before rotoscoping helps an artist predict where the shape needs to move next, place keyframes efficiently, and spot moments where the subject changes direction or speed, which is exactly where extra keyframes are needed.

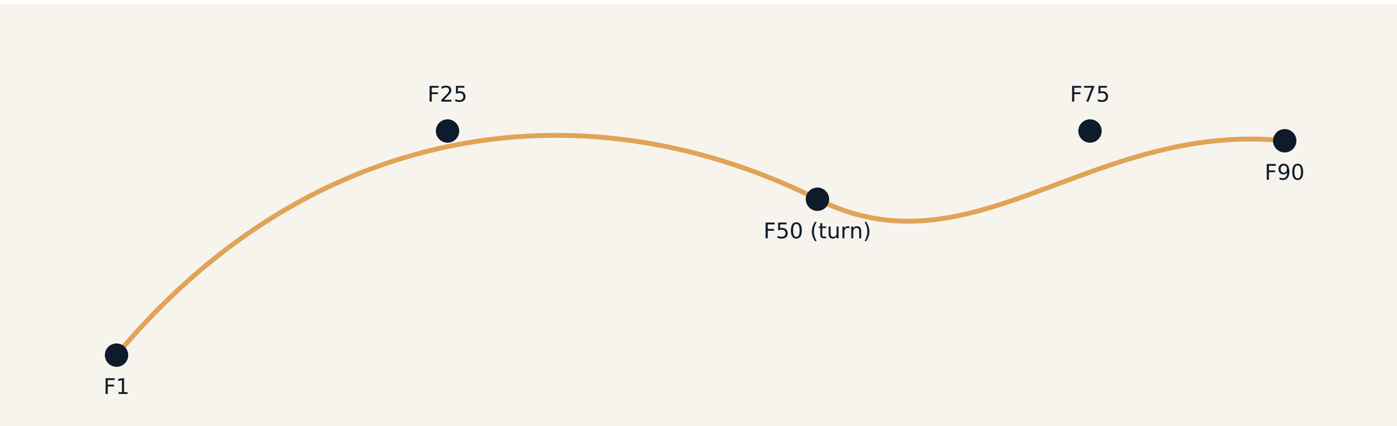

Fig 7.2 — A motion path traced across a shot; note the extra keyframe placed at Frame 50 exactly where the subject changes direction.

Combining the Two Concepts in Practice

- First, identify the subject's positive space silhouette clearly, including any interior negative-space gaps.
- Then, study the motion path the subject will take across the whole shot before placing a single keyframe — watch the footage in full first.
- Place keyframes at the start, end, and every point where the motion path changes direction or speed — not at even, arbitrary intervals.
- Let the software interpolate the smooth sections of the path between those key changes.

WHY THIS MATTERS

A roto artist who understands the motion path in advance places far fewer, far smarter keyframes — resulting in a shape that tracks more accurately with less manual correction, because keyframes sit exactly where the software actually needs guidance.

WORKED EXAMPLE

A dancer spins, pauses, then leaps sideways. Instead of setting a keyframe every 5 frames blindly, the artist watches the full clip first, then places keyframes at: the start of the spin, the pause, the beginning of the leap, and the landing — four purposeful keyframes instead of twenty arbitrary ones, with smoother results.

KEY TAKEAWAYS

- Positive space is the subject; negative space is everything else, including interior gaps that must be excluded from the matte.
- A motion path is the trajectory a shape follows across a shot's timeline.
- Keyframes belong at direction/speed changes along the motion path, not at arbitrary intervals.
- Studying the full shot before rotoscoping leads to fewer, smarter keyframes.

Keyframing Techniques Used in Rotoscopy

Breakdown keyframes, spacing strategy, and interpolation — the craft of animating a roto shape efficiently.

What is a Keyframe in Rotoscopy?

A keyframe is a specific frame where the exact position and shape of every point on a spline is explicitly recorded. Between two keyframes, the software mathematically calculates ("interpolates") the in-between positions automatically. Good keyframing is the single biggest factor in how much manual correction a roto shot needs.

Core Keyframing Techniques

1. Bookend Keyframing

Set a keyframe at the very first and very last frame of the shape's motion first, establishing the overall range before adding any detail in between.

2. Breakdown Keyframes

After the bookend keyframes are set, the artist scrubs to the frame where the interpolation drifts furthest off the real edge — usually the midpoint of the motion — and adds a "breakdown" keyframe there to correct it. This process repeats, splitting the remaining gaps in half, until the shape tracks accurately throughout.

3. Tracker-Assisted Keyframing

Rather than manually keying every correction, the artist runs the shape through an automatic point or planar tracker first, then only adds manual keyframes at frames where the tracker visibly loses the edge (due to occlusion, motion blur, or low contrast).

4. Sparse Keyframing for Efficiency

Professional artists deliberately use as few keyframes as the shot allows, trusting the software's interpolation for smooth, predictable motion, and reserving manual keyframes only for genuine direction or speed changes.

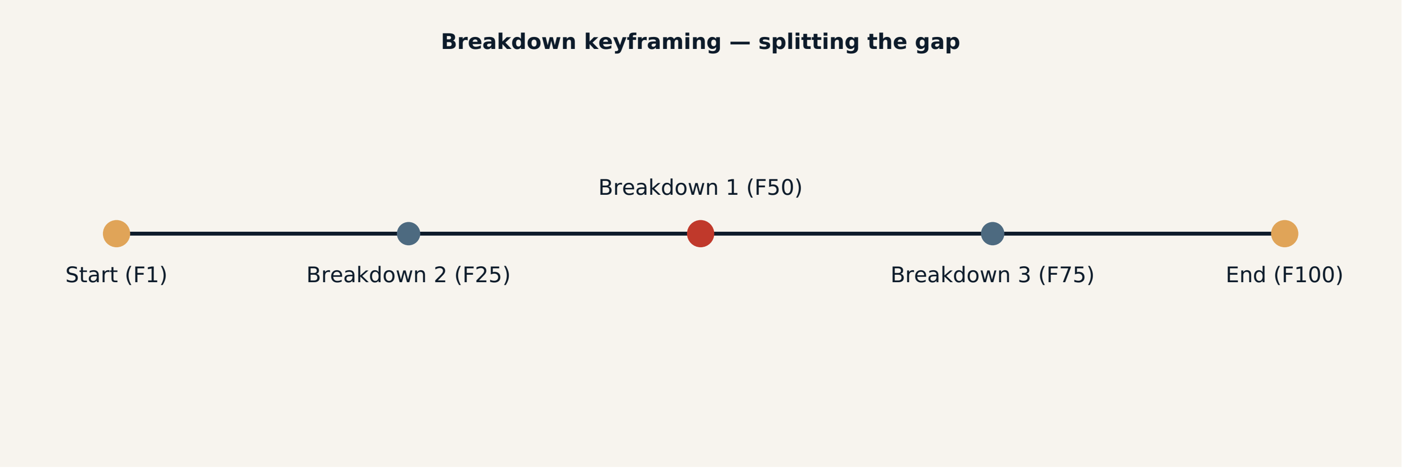

Fig 8.1 — Breakdown keyframing: start with two bookend keys, then repeatedly halve the largest remaining gap.

Interpolation Types

TYPE	BEHAVIOUR	BEST USED FOR
Linear	Constant, mechanical speed between keyframes	Rigid, machine-like or uniform-speed motion
Smooth / Spline	Eased acceleration and deceleration	Organic human/animal motion

Hold / Stepped	No interpolation — value snaps at each keyframe	Matching sudden, snap-like movement or cuts

COMMON MISTAKE

Setting keyframes at evenly-spaced intervals (every 10 frames, for example) regardless of the actual motion. This wastes time on frames that didn't need correction and under-corrects the frames that did — always key based on where the motion actually changes, not the clock.

WORKED EXAMPLE

Rotoscoping a bouncing ball, the artist sets bookend keyframes at the top of the first bounce and the top of the last bounce, then adds breakdown keyframes only at each moment the ball touches the ground (where direction reverses) — five purposeful keyframes capture ten seconds of bouncing accurately, instead of fifty evenly-spaced ones.

KEY TAKEAWAYS

- Bookend keyframes define the range; breakdown keyframes correct drift at the points of greatest error.
- Trackers can automate most of the in-between work, leaving manual keys only for tracker failures.
- Sparse, purposeful keyframing beats evenly-spaced keyframing every time.
- Choose linear, smooth or stepped interpolation to match the true nature of the subject's motion.

CHAPTER 09

Motion-Based Rotoscopy & Linear Movement

Adapting roto strategy to subjects that move at a constant, predictable speed and direction.

What is Motion-Based Rotoscopy?

Motion-based rotoscopy is an approach where the artist builds their keyframing and tracking strategy specifically around the *type* of motion the subject performs, rather than applying one generic method to every shot. A subject that moves in a straight line at constant speed (linear movement) is rotoscoped very differently from one that accelerates, curves, or changes direction unpredictably.

Linear Movement

Linear movement describes motion along a straight trajectory at a constant or near-constant speed — a car driving straight down a road, a train passing through frame, or a package sliding along a conveyor belt. Because the position of the subject at any frame can be predicted mathematically from just two points, linear movement is the easiest category of motion to rotoscope efficiently.

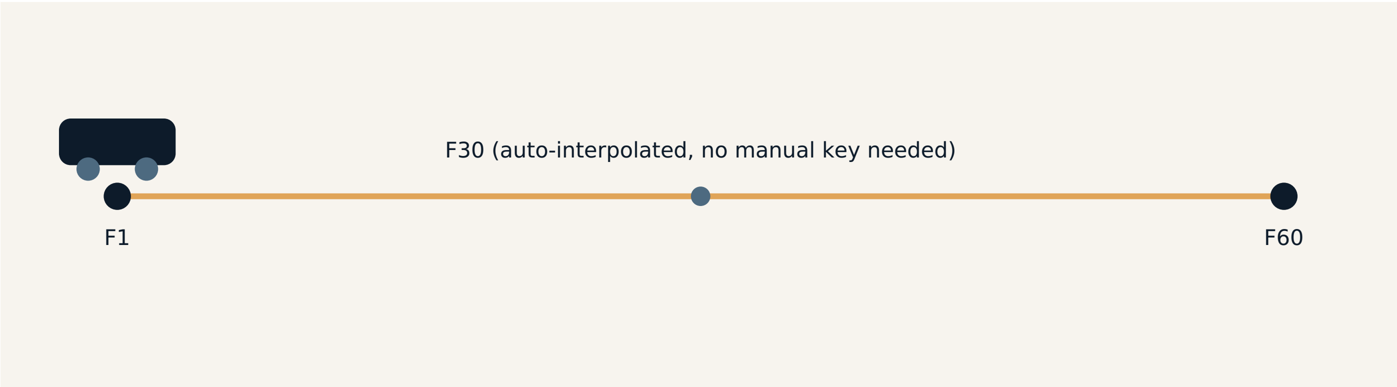

Fig 9.1 — Pure linear movement needs only two keyframes (start and end); every in-between frame can be trusted to linear interpolation.

The Two-Keyframe Strategy

For genuinely linear motion, an artist only needs to set the shape correctly at the very first frame and the very last frame of the movement, then set the interpolation type to **Linear** (not Smooth). The software will calculate every in-between frame along a perfectly straight, constant-speed path — matching the real footage almost exactly, with zero wasted breakdown keyframes.

CAUTION

Real-world motion is rarely *perfectly* linear — a "constant speed" car may still wobble slightly on an uneven road. Always scrub through the two-keyframe range and check for small drift; add a light correction keyframe only if genuine drift appears.

Combining Linear Segments for Complex Motion

Many real shots are not *entirely* linear but contain distinct linear segments joined by a change in direction or speed — for example, a skateboarder rolling in a straight line, then turning sharply. In such cases, the artist treats each straight segment as its own two-keyframe linear block, and places a single breakdown keyframe exactly at the moment direction changes, effectively chaining several efficient linear sections together instead of over-keying the whole shot.

WORKED EXAMPLE

A shot shows a delivery drone flying in a straight line across the frame for 80 frames before banking to change direction.

The artist places keyframes at Frame 1 and Frame 80 with linear interpolation for the straight flight, then adds one breakdown keyframe at the exact bank point (Frame 80) and a new end keyframe for the second straight segment — covering the entire 150-frame shot with only three keyframes.

KEY TAKEAWAYS

- Motion-based rotoscopy tailors the keyframing strategy to the actual type of motion in the shot.
- Linear movement — straight-line, constant speed — needs as few as two keyframes with linear interpolation.
- Always verify true linearity by scrubbing; real motion often has small deviations.
- Complex shots can be broken into a chain of efficient linear segments joined at direction-change points.

CHAPTER 10

Handling Blurred Edges & Motion Blur in Rotoscopy

Rotoscoping subjects whose edges are naturally soft because of fast movement or a shallow camera focus.

Why Blurred Edges Are Difficult

When a subject moves quickly during a single frame's exposure time, its edge smears across several pixels instead of forming one crisp line — this is **motion blur**. A shape drawn with a hard, sharp edge around a naturally blurred subject will look pasted-on and artificial, because real motion blur has a soft, semi-transparent gradient rather than a clean cut-off.

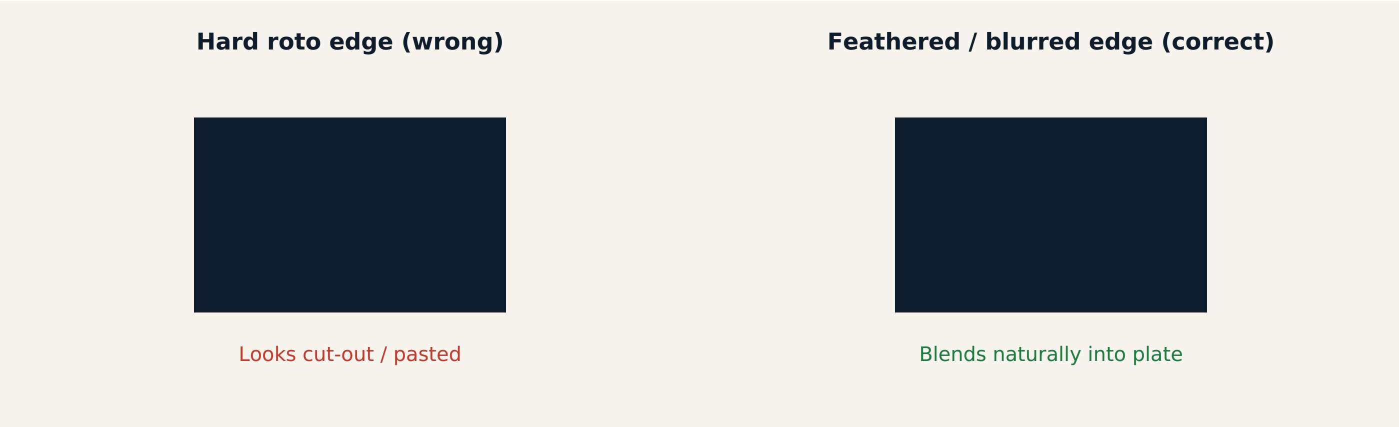

Fig 10.1 — A hard-edged matte reads as artificial; matching the natural motion-blur gradient makes the composite believable.

Techniques for Handling Motion Blur

1. Enable the Shape's Motion Blur Setting

Most professional roto tools include a per-shape motion-blur toggle that automatically softens the matte edge along the direction of travel, based on the shape's own speed between keyframes — this is the fastest, most accurate starting point.

2. Match Shutter Angle

Motion blur amount is tied to the camera's shutter angle at the time of the shoot (commonly 180°). If this value is known or can be estimated from the footage, matching it in the software's motion-blur settings produces a far more accurate blur profile than a guess.

3. Use Variable Feather Instead of a Single Uniform Value

Real motion blur is rarely equal on every side of a subject — the leading and trailing edges in the direction of movement blur more than the edges perpendicular to the motion. Advanced tools allow "per-point" feather so the leading/trailing edges get more softness than the top/bottom edges.

4. Roto Slightly Inside the Blurred Region, Then Feather Outward

Rather than trying to trace the exact outer limit of a blur trail (which is often too faint and inconsistent to track reliably), many artists draw the shape at the point where the blur is roughly 50% opacity, then add feather to extend softly outward to the true edge.

5. Sample and Compare Against the Plate

After adding motion blur/feather, the artist should zoom in and compare the composited edge directly against the original plate's blur trail, frame by frame, adjusting until the two are indistinguishable.

COMMON MISTAKE

Applying a single, fixed feather value to an entire shot regardless of how fast the subject is moving at each moment — a fast-moving frame needs more blur/feather than a nearly-still frame of the same shot, so this value should often be animated.

WORKED EXAMPLE

A roto artist works on a shot of a boxer's glove striking forward. During the strike itself (a handful of frames), the glove is heavily motion-blurred, so the artist enables shape motion blur and increases feather sharply for just those frames; before and after the strike, when the glove is nearly still, feather is reduced back to a normal, tight value — matching the varying blur seen in the plate throughout.

KEY TAKEAWAYS

- Motion blur makes a moving subject's true edge soft and semi-transparent, not sharp.
- Enable per-shape motion blur and, where possible, match the original shutter angle.
- Use variable/per-point feather rather than one uniform value across the whole shape.
- Animate feather to increase during fast motion and decrease when the subject is still.
- Always compare the finished edge directly against the original blurred plate before finalising.

QUICK REFERENCE

Glossary at a Glance

A condensed reference of every core term covered across these notes.

TERM	SHORT DEFINITION
Rotoscopy	Frame-by-frame tracing/masking of a subject using vector shapes.
Keying	Automatic background removal based on colour or brightness (chroma/luma key).
Spline	A smooth vector curve made from connected points.
Bezier / B-Spline / X-Spline	The three main curve types, differing in how handles and points control the curve.
Matte	The black-and-white mask produced by a roto shape.
Garbage Matte / Core Matte	A rough exclusion shape vs. a tightly-fit solid-opacity shape.
Feather	Soft edge falloff applied to a matte for natural blending.
Positive / Negative Space	The subject's silhouette vs. the surrounding/interior empty area.
Motion Path	The trajectory a shape follows across a shot's frames.
Keyframe / Breakdown Keyframe	An explicitly recorded shape position; a corrective key added between bookends.
Interpolation	How the software calculates in-between frames (linear, smooth, stepped).
Edge Consistency	How stable and believable the matte edge remains across every frame.
Motion Blur (roto)	Matching the natural soft blur of a fast-moving subject's edge.
Linear Movement	Straight-line, constant-speed motion needing minimal keyframes.

HOW TO USE THESE NOTES

- Read each chapter alongside the software you are practising on — pause and try each numbered step live.
- Revisit the diagrams (figures) whenever a term feels abstract; they are designed to mirror what you'll actually see on screen.
- The worked examples show how each concept behaves on a real, practical shot — try recreating a similar mini-exercise yourself.